
T1 MRI Denoising with a 2D Residual U-Net

Computational Neuroscience Final Project

T1 MRI Denoising Task

T1_noisy.nii.gz $\rightarrow$ T1_clean.nii.gz

Abstract

Recovering clean anatomical structure from noisy magnetic resonance images is a practical image restoration problem with direct relevance to medical image analysis. This work evaluates a supervised denoising pipeline for paired T1 MRI volumes, where each noisy scan is matched with a clean target from the same subject. We train a 2D Residual U-Net on axial slices and use case-level data splits to prevent subject leakage. The full experiment includes 600 paired cases, divided into 420 training, 90 validation, and 90 test cases. On the held-out test set, the model reduces MAE from 0.0044 to 0.0028 and MSE from 3.72×10^{-5} to 1.80×10^{-5} relative to the noisy baseline. It also improves PSNR from 44.7757 dB to 47.7168 dB and SSIM from 0.9470 to 0.9930, with PSNR gains in all 90 test cases. These results show that residual slice-wise learning provides a simple, stable, and reproducible baseline for T1 MRI denoising.

1 Introduction

MRI denoising is a core image restoration problem. Given a noisy image x and a clean target image y , the goal is to learn a mapping $f_{\theta}(x)$ that suppresses noise while preserving anatomical structure. This work focuses on paired T1 denoising because the input and target images belong to the same modality, have aligned spatial structure, and allow direct supervised training with standard restoration metrics.

We implement a 2D Residual U-Net baseline. U-Net architectures are widely used in biomedical image analysis because the encoder-decoder structure captures multi-scale context while skip connections preserve spatial detail [Ronneberger et al., 2015]. For denoising, residual prediction is natural: noisy and clean images share most anatomical content, so the network mainly needs to estimate a correction term rather than synthesize a full image from scratch. The experiment evaluates whether this baseline improves pixel-level error and structural similarity over the noisy input.

2 Data

The data root is `data/cn_project_t1_noise2`. Each case directory contains:

- `T1_noisy.nii.gz`: the noisy T1 MRI volume;
- `T1_clean.nii.gz`: the paired clean T1 MRI volume.

The full experiment includes 600 complete paired cases. We split the data at the case level, using a fixed random seed of 42. The resulting train, validation, and test sets contain 420, 90, and 90 cases, respectively. Case-level splitting is important because slice-level random splitting could place different slices from the same subject into both training and test sets, causing data leakage.

3 Method

3.1 Preprocessing

For each case, the noisy and clean NIfTI volumes are loaded as 3D arrays. Intensities are normalized per case by computing the 1st and 99th percentiles over the joint foreground voxels from both noisy and clean images, then clipping and scaling values to $[0, 1]$. We extract axial 2D slices and skip near-empty slices. Each slice is resized to 256×256 before being passed to the model. Evaluation metrics are computed on the same normalized and resized 2D slices used by the model.

3.2 Residual U-Net

The model is a lightweight 2D U-Net with one-channel input and output. The encoder uses convolutional blocks and max pooling; the decoder uses transposed convolutions and skip connections. Instead of predicting the denoised image directly, the network predicts a residual $r_\theta(x)$:

$$\hat{y} = \text{clip}(x + r_\theta(x), 0, 1). \quad (1)$$

The final residual convolution is initialized to zero, so the initial model behaves like the noisy baseline. This makes training stable because the network starts from an identity-like solution and learns only the denoising correction.

3.3 Training objective

The loss combines L1 and MSE terms:

$$\mathcal{L}(\hat{y}, y) = \|\hat{y} - y\|_1 + 0.2\|\hat{y} - y\|_2^2. \quad (2)$$

The L1 term encourages robust pixel-wise reconstruction, while the MSE term penalizes larger deviations. We use AdamW [Kingma and Ba, 2015] with learning rate 10^{-4} and weight decay 10^{-5} . The model is trained for 40 epochs with batch size 24 on an NVIDIA RTX A6000 GPU. The checkpoint with the lowest validation loss is selected for final evaluation.

4 Evaluation

We compare the trained model against the noisy input baseline on the held-out test set. The metrics are mean absolute error (MAE), mean squared error (MSE), peak signal-to-noise ratio (PSNR), and structural similarity index (SSIM) [Wang et al., 2004]. MAE and MSE are better when lower; PSNR and SSIM are better when higher. All reported values are case-level means over the 90 test cases.

5 Results

5.1 Training behavior

The training loss decreases from 0.003040 at epoch 1 to 0.002538 at epoch 40. The best validation loss is 0.002698 and occurs at epoch 36. Figure 1 shows the training and validation curves.

5.2 Quantitative results

Table 1 summarizes the test-set results. The Residual U-Net improves every metric over the noisy baseline. It reduces MAE by 0.0016, reduces MSE by 1.92×10^{-5} , improves PSNR by 2.9411 dB, and improves SSIM by 0.0460. At the case level, the model achieves higher PSNR than the noisy baseline for all 90 test cases.

Table 2 lists the first 10 test cases from the case-level metric file. The full file is available at `runs/denoise_unet/eval/case_metrics.csv`.

5.3 Qualitative results

Figure 2 shows three representative test cases. Each panel contains the noisy slice, denoised output, clean target, and absolute error map. The denoised outputs are visually closer to the clean targets, and the error maps are weaker after denoising.

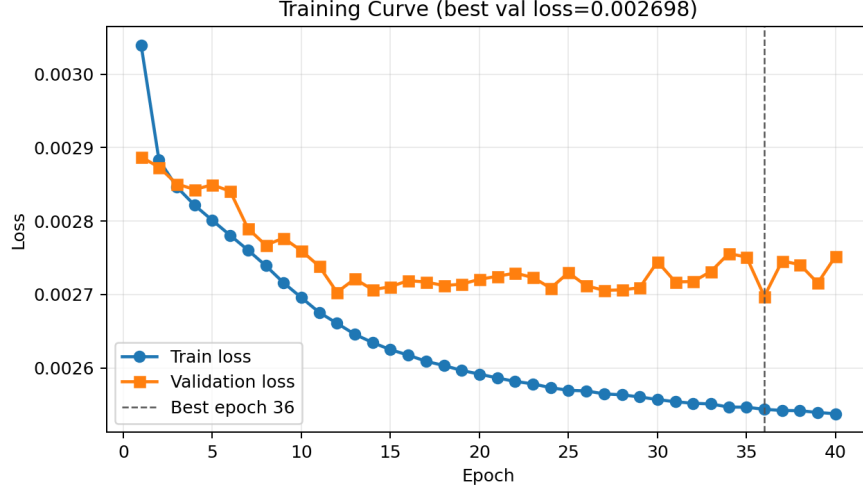


Figure 1: Training and validation loss curves. The best validation checkpoint occurs at epoch 36.

Table 1: Case-level mean metrics on the 90-case test set.

Method	MAE ↓	MSE ↓	PSNR ↑	SSIM ↑
Noisy baseline	0.0044	3.72×10^{-5}	44.7757	0.9470
Residual U-Net	0.0028	1.80×10^{-5}	47.7168	0.9930

6 Discussion

The results show that a 2D Residual U-Net is a strong baseline for this paired T1 denoising task. The model consistently improves pixel-wise error and structural similarity compared with the noisy input. The large SSIM gain suggests that the model does not merely smooth the image, but also preserves the main anatomical structures.

There are several limitations. First, the model processes axial slices independently and does not explicitly use 3D context from adjacent slices. A 2.5D model with neighboring slices as channels, or a 3D U-Net, may further improve consistency across slices. Second, the evaluation is performed on normalized resized 2D slices rather than original-resolution 3D volumes. Third, optional engineering improvements such as early stopping, mixed precision, and data augmentation are implemented in the codebase but are not part of the reported full-run results; using them in the final result would require retraining and updating the metrics.

7 Conclusion

This project implements a complete supervised T1 MRI denoising experiment. On 90 held-out test cases, the Residual U-Net improves MAE, MSE, PSNR, and SSIM over the noisy baseline, with PSNR improvement in all test cases. The experiment is simple, reproducible, and suitable as a course final-project baseline for medical image denoising.

Code availability

The code and report files are available at https://github.com/liuxiangyu666777888/Computational_Neurobiology.

Table 2: Case-level results for the first 10 test cases.

Case	Slices	Noisy MAE	Model MAE	Noisy PSNR	Model PSNR	Noisy SSIM	Model SSIM
1000542	206	0.0045	0.0024	44.4386	48.5405	0.9327	0.9954
1002722	206	0.0031	0.0023	47.5008	49.3120	0.9791	0.9930
1002807	167	0.0060	0.0035	42.0188	45.5105	0.9217	0.9936
1003444	198	0.0046	0.0028	44.0850	47.3513	0.9429	0.9953
1003618	211	0.0027	0.0022	48.7816	49.9918	0.9862	0.9939
1005201	189	0.0036	0.0022	46.3301	49.5584	0.9579	0.9965
1005430	184	0.0044	0.0027	44.6042	47.8435	0.9513	0.9958
1008080	197	0.0058	0.0032	42.2767	46.2604	0.9057	0.9923
1009859	206	0.0042	0.0028	44.7305	47.3590	0.9621	0.9923
1010191	215	0.0038	0.0026	45.7588	48.0494	0.9711	0.9922

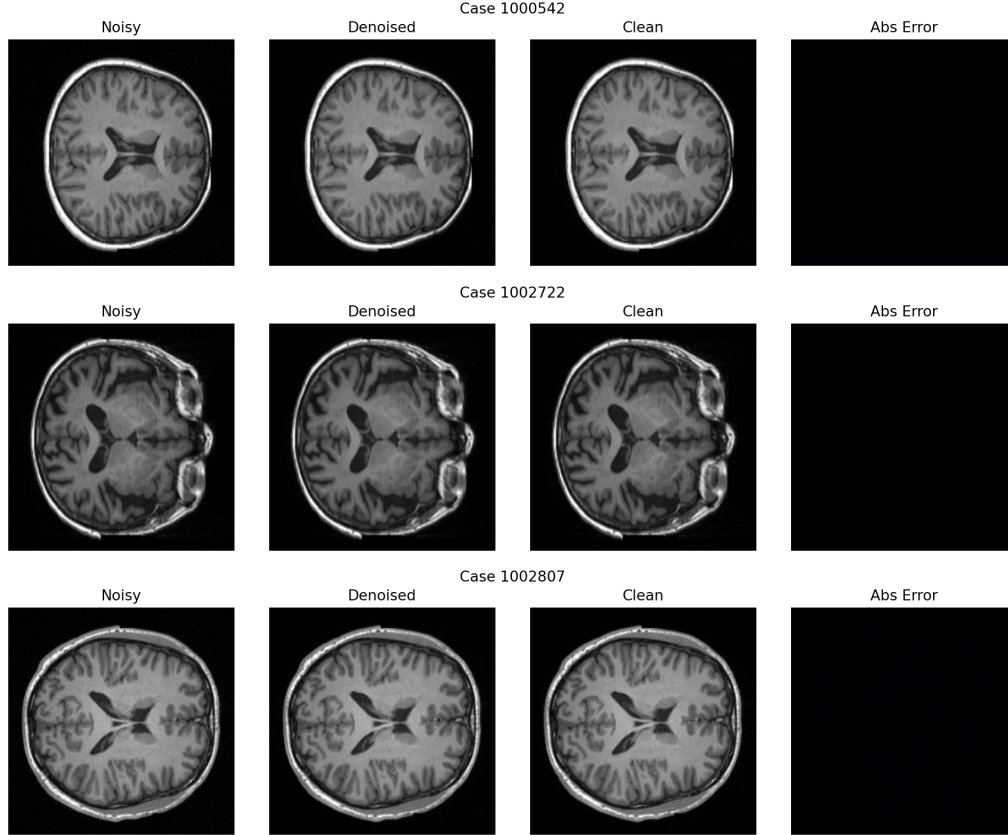


Figure 2: Qualitative results for three test cases. In each row, the columns are Noisy, Denoised, Clean, and Absolute Error.

References

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- Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention*, pages 234–241. Springer, 2015.
- Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4): 600–612, 2004. doi: 10.1109/TIP.2003.819861.